

HILO 5 S, HI, USA

WMO: 912960

Lat: 19.6450N

Lon: 155.0827W

Elev: 190

StdP: 99.07

Time Zone: -10.00 (W10)

Period: 05-19

WBAN: 21515

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.4	15.2	12.0	8.9	17.1	12.9	9.5	17.5	3.1	25.2	3.0	24.9	0.1	N/A	0.187

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.7	27.9	22.6	27.3	22.2	26.8	21.9	24.1	26.4	23.6	25.9	23.1	25.3	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.3	18.5	25.3	22.9	18.0	24.9	22.4	17.5	24.5	73.2	26.5	71.4	25.8	69.6	25.5	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2.9	2.5	2.2	12.8	29.2	0.5	1.0	12.5	29.9	12.2	30.5	11.9	31.1	11.6	31.9
			11.5	24.9	0.5	0.9	11.2	25.5	10.8	26.0	10.5	26.5	10.2	27.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.4	20.0	19.9	20.1	20.7	21.4	21.9	22.4	22.8	22.6	22.3	21.4	20.6
	DBStd	1.49	1.07	1.41	1.26	1.13	1.02	0.82	1.01	1.02	1.07	1.03	1.11	1.10
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	5	1	2	2	0	0	0	0	0	0	0	0	0
	CDD10.0	4143	310	278	314	321	352	356	385	397	379	381	342	327
	CDD18.3	1107	53	47	58	71	94	106	127	138	129	123	92	69
	CDH23.3	3281	114	121	122	138	250	297	458	505	534	404	219	117
CDH26.7	137	2	6	4	3	7	7	17	28	37	18	6	2	
Wind	WSAvg	0.6	0.6	0.7	0.7	0.7	0.7	0.7	0.7	0.6	0.6	0.5	0.5	0.6
Precipitation	PrecAvg	3683	276	329	393	310	242	214	280	276	259	296	424	383
	PrecMax	6033	824	927	1055	579	673	688	596	584	627	647	1164	749
	PrecMin	1946	11	36	35	127	60	79	80	111	48	85	35	112
	PrecStd	746	198	211	248	106	130	114	118	111	146	127	256	178
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	27.5	27.1	27.0	27.5	27.6	28.0	28.6	28.9	28.3	27.5	26.8
		MCWB	20.1	20.4	21.0	21.1	21.4	21.8	22.9	23.6	23.5	22.9	22.5	21.0
	2%	DB	25.7	26.0	25.8	25.7	26.4	26.5	27.1	27.5	27.7	27.1	26.4	25.5
		MCWB	19.9	20.5	20.5	20.5	21.0	21.4	22.3	22.9	22.9	22.5	21.8	20.8
	5%	DB	24.7	24.9	24.8	24.9	25.6	25.7	26.5	26.7	26.9	26.4	25.5	24.7
		MCWB	19.7	20.0	20.2	20.2	20.7	21.1	21.9	22.4	22.5	22.2	21.3	20.5
	10%	DB	23.7	23.7	23.7	24.1	24.9	25.1	25.8	25.9	26.2	25.6	24.7	23.9
		MCWB	19.4	19.6	19.8	20.0	20.5	20.9	21.7	22.1	22.1	21.9	21.0	20.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	22.2	22.1	22.4	22.5	22.9	24.2	24.7	24.8	24.2	23.0	22.6
		MCDB	24.7	25.1	25.4	25.3	25.3	25.6	26.6	27.2	27.8	26.3	25.7	25.0
	2%	WB	20.9	21.4	21.4	21.6	21.9	22.3	23.5	24.0	24.0	23.5	22.6	21.7
		MCDB	23.9	24.3	24.5	24.0	24.7	25.0	25.7	26.2	26.4	25.7	25.0	23.9
	5%	WB	20.5	20.8	20.8	21.0	21.5	21.9	22.9	23.5	23.5	23.0	22.1	21.2
		MCDB	23.4	23.8	23.5	23.5	24.2	24.5	25.1	25.5	25.9	25.3	24.4	23.3
	10%	WB	20.1	20.3	20.3	20.6	21.0	21.5	22.4	23.0	22.9	22.5	21.6	20.6
		MCDB	22.8	23.1	22.9	23.2	23.7	24.1	24.7	24.9	25.3	24.8	23.8	22.7
Mean Daily Temperature Range	5% DB	MDBR	7.8	7.4	6.9	6.7	7.1	6.9	6.9	6.7	7.4	7.0	6.8	6.8
		MCDBR	9.3	9.4	8.9	8.1	8.2	7.9	7.8	7.7	8.2	7.9	7.9	8.1
		MCWBR	5.1	4.9	4.7	3.8	4.1	3.8	3.7	3.8	4.3	4.4	4.5	4.6
	5% WB	MCDBR	8.0	7.7	8.0	6.8	7.1	6.7	6.6	6.1	7.0	6.9	6.8	6.4
		MCWBR	4.9	4.4	4.6	3.7	4.0	3.7	3.8	3.7	4.2	4.2	4.3	4.2
Clear-Sky Solar Irradiance	taub	0.359	0.361	0.378	0.416	0.397	0.385	0.391	0.377	0.375	0.373	0.369	0.354	
	taud	2.575	2.568	2.517	2.373	2.463	2.488	2.482	2.534	2.538	2.569	2.582	2.613	
	Ebn at Noon	922	938	930	891	897	899	896	915	919	914	905	917	
	Edh at Noon	94	99	108	126	114	110	111	106	104	98	92	88	
All-Sky Solar Radiation	RadAvg	4.49	4.87	5.33	5.87	6.16	6.33	6.44	6.26	5.83	5.10	4.38	4.07	
	RadStd	0.27	0.33	0.58	0.40	0.39	0.23	0.29	0.30	0.24	0.20	0.25	0.39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+125	+125

Nomenclature: See separate page